

Calibrating the Heston Stochastic Volatility Model to S&P 500 Index Options: A Practitioner's Approach

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1 Introduction

In the Black-Scholes (1973) framework, the volatility of an asset is assumed to be constant over time. While convenient, this assumption is systematically violated by market data: options with different strikes but the same expiry date imply different levels of volatility when their prices are inverted through the Black-Scholes formula. This phenomenon, known as the volatility smile or volatility skew, is one of the most well-documented stylised facts in derivatives markets.

This project adopts the perspective of a quantitative finance practitioner working on an equity derivatives desk. The central objective is to calibrate the Heston (1993) stochastic volatility model to real market data from S&P 500 (SPX) index options, and to evaluate how accurately the model reproduces the observed implied volatility surface.

The practical motivation is clear: any desk that prices or hedges equity options must maintain a model that is consistent with market-quoted implied volatilities across strikes and maturities. Black-Scholes with a flat volatility fails this requirement. Heston, by allowing volatility to be stochastic and mean-reverting, provides a framework capable of generating realistic volatility surfaces.

The project proceeds as follows. Section 2 provides background on implied volatility, the volatility smile, and existing approaches to stochastic volatility modelling. Section 3 derives the Heston model in detail, including its stochastic differential equations, characteristic function, and pricing formula. Section 4 describes the option data used. Section 5 explains the calibration procedure. Section 6 presents and interprets the results. Section 7 concludes.

2 Background

2.1 Implied Volatility and the Volatility Smile

The Black-Scholes formula prices a European call option as:

$$C^{BS}(S_0, K, T, r, \sigma) = S_0 \Phi(d_1) - Ke^{-rT} \Phi(d_2), \quad (1)$$

where $d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$, $d_2 = d_1 - \sigma\sqrt{T}$, and $\Phi(\cdot)$ is the standard normal CDF.

The implied volatility $\sigma_{IV}(K, T)$ of an option with strike K and maturity T is the value of σ that equates the Black-Scholes formula to the observed market price:

$$C^{\text{mkt}}(K, T) = C^{BS}(S_0, K, T, r, \sigma_{IV}(K, T)). \quad (2)$$

Since there is no closed-form inverse, σ_{IV} is found numerically via root-finding, here using

Brent's method.

If Black-Scholes were a perfect model, σ_{IV} would be the same for all (K, T) . In practice, for equity indices, σ_{IV} is higher for low strikes, corresponding to out-of-the-money puts, and decreases as the strike increases. This pattern is called the volatility skew. The volatility surface $\sigma_{IV}(K, T)$ therefore describes a two-dimensional function that any calibrated model should reproduce.

2.2 The Leverage Effect

A key driver of the equity-index skew is the leverage effect: when the market falls sharply, implied volatility tends to increase, while during rallies volatility is usually more subdued. In the Heston model this asymmetry is mainly controlled by the correlation parameter ρ . A negative value of ρ means that negative shocks to the index tend to coincide with positive shocks to variance, generating the observed left skew.

2.3 Stochastic Volatility Models

Several model families have been proposed to generate a non-flat volatility surface. Hull and White (1987) introduced an early stochastic volatility framework; Stein and Stein (1991) specified an Ornstein-Uhlenbeck process for volatility. The GARCH model of Engle (1982) and Bollerslev (1986) achieves a similar objective in discrete time: tomorrow's variance depends on today's variance and today's shock. Heston (1993) provides a continuous-time model that has become a standard benchmark in equity derivatives because it admits a semi-analytical pricing formula through the characteristic function.

3 The Heston Model

3.1 Model Dynamics

Under the risk-neutral measure $\mathbb{Q}$, the Heston model specifies the joint evolution of the asset price S_t and its instantaneous variance v_t as the following system of stochastic differential equations:

$$dS_t = r S_t dt + \sqrt{v_t} S_t dW_t^S, \tag{3}$$

$$dv_t = \kappa(\theta - v_t) dt + \sigma \sqrt{v_t} dW_t^v, \tag{4}$$

with $\text{corr}(dW_t^S, dW_t^v) = \rho dt$.

The five parameters of the model are:

Parameter	Economic meaning	Typical range (equity)
v_0	Initial variance	(0, 0.50)
κ	Speed of mean reversion	(0, 10)
θ	Long-run variance	(0, 0.50)
σ	Volatility of variance, or vol-of-vol	(0, 3)
ρ	Spot-variance correlation	(-1, 0)

Equation (4) is a Cox-Ingersoll-Ross (CIR) process: v_t reverts to its long-run level θ at speed κ , and the diffusion coefficient $\sigma\sqrt{v_t}$ makes the volatility of variance state dependent. The Feller condition, $2\kappa\theta \geq \sigma^2$, is a sufficient condition for v_t to remain strictly positive. In empirical calibrations this condition is often violated, especially for short-dated option data, so in this project it is reported but not imposed as a hard constraint.

The correlation ρ is the key parameter that generates the volatility skew. When $\rho < 0$, a downward move in S_t tends to coincide with an upward move in v_t , reproducing the leverage effect and, through option prices, the observed left skew.

3.2 Relationship to GARCH

The Heston model can be interpreted as a continuous-time analogue of the GARCH(1,1) model. GARCH(1,1) specification,

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (5)$$

describes variance as a weighted average of its own lag and the squared innovation. The key difference is that Heston operates in continuous time and is therefore directly compatible with option pricing under the risk-neutral measure.

3.3 The Characteristic Function

Heston's key insight was that the characteristic function of $\ln S_T$ under $\mathbb{Q}$ admits a closed-form expression. We use the numerically stable form :

$$\varphi(\phi) = \mathbb{E}^{\mathbb{Q}}[e^{i\phi \ln S_T}] = e^{C(\phi, T) + D(\phi, T) v_0 + i\phi \ln S_0}, \quad (6)$$

where

$$d = \sqrt{(\rho\sigma i\phi - \kappa)^2 + \sigma^2(i\phi + \phi^2)}, \quad (7)$$

$$g = \frac{\kappa - \rho\sigma i\phi + d}{\kappa - \rho\sigma i\phi - d}, \quad (8)$$

$$C(\phi, T) = i\phi rT + \frac{\kappa\theta}{\sigma^2} \left[(\kappa - \rho\sigma i\phi + d) T - 2 \ln \left(\frac{1 - g e^{dT}}{1 - g} \right) \right], \quad (9)$$

$$D(\phi, T) = \frac{\kappa - \rho\sigma i\phi + d}{\sigma^2} \cdot \frac{1 - e^{dT}}{1 - g e^{dT}}. \quad (10)$$

3.4 Option Pricing

A European call price can be computed from the characteristic function using the following single-integral representation:

$$C(S_0, K, T, r) = \frac{S_0 - K e^{-rT}}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{e^{rT} \varphi(\phi - i) - K \varphi(\phi)}{i\phi K^{i\phi}} \right] d\phi. \quad (11)$$

In the Python implementation, the integral is evaluated using the midpoint rule over the finite interval $[0, 100]$ with $N = 300$ integration subintervals. This choice is a compromise between numerical accuracy and computation time. Put prices are obtained from call prices using put-call parity:

$$P(S_0, K, T, r) = C(S_0, K, T, r) - S_0 + K e^{-rT}. \quad (12)$$

4 Data

4.1 Source and Collection

Option data were downloaded from Yahoo Finance via the `yfinance` Python library on 26 April 2026, when the S&P 500 index stood at $S_0 = 7,165.08$. Five expiry dates were initially selected to cover very short to one-month maturities. The shortest available maturity, approximately three calendar days, was excluded from the final calibration because such ultra-short-dated options are particularly sensitive to market microstructure noise and are difficult for a constant-parameter diffusion model to fit.

The four maturities retained for calibration were:

Expiry date	T (years)	Approx. calendar days
2026-05-04	0.022	8
2026-05-11	0.041	15
2026-05-18	0.060	22
2026-05-26	0.082	30

The risk-free rate was proxied by the 3-month US Treasury bill yield, $r = 3.59\%$, obtained from Yahoo Finance (`^IRX`). For simplicity, no dividend yield or forward-price adjustment is used. This is discussed as a limitation in Section 7.

4.2 Cleaning and Filtering

The raw option chain contained 1,270 records. The following filters were applied sequentially:

1. Remove options with missing bid, ask, volume, or open interest.
2. Remove options with bid = 0.
3. Require volume ≥ 10 and open interest ≥ 10 .
4. Restrict to moneyness $K/S_0 \in [0.80, 1.20]$.
5. Remove options with ask/bid > 3 .
6. Exclude maturities below 0.02 years.

After filtering, 615 options remained before selecting puts only. The mid-price $p^{\text{mid}} = (\text{bid} + \text{ask})/2$ was used as the market price for each option.

4.3 Implied Volatility Computation

The implied volatility of each option was computed by inverting the Black-Scholes formula numerically using Brent’s method. Options for which no solution existed within $\sigma \in [0.01, 5.0]$ were discarded. The resulting implied volatility statistics for the retained put sample are reported in Table 1.

Table 1: Summary statistics of market implied volatilities for the retained put sample.

	Count	Mean	Std	Min	25%	Median	Max
IV	340	0.224	0.094	0.120	0.168	0.197	0.729

The highest implied volatilities occur for low-moneyness put options and reflect the market price of downside protection.

5 Methodology

5.1 Calibration on Put Options

The Heston model is calibrated on put options only. For equity indices, out-of-the-money puts are typically important because institutional investors use them as portfolio insurance. As a result, put quotes are especially informative about downside risk and the left skew, which are precisely the features that the Heston model is designed to capture.

5.2 Objective Function

Calibration is performed by minimising a weighted mean-squared error on implied volatilities:

$$\mathcal{L}(v_0, \kappa, \theta, \sigma, \rho) = \frac{1}{N} \sum_{j=1}^N w_j \left[\sigma_{IV}^{\text{model}}(K_j, T_j) - \sigma_{IV}^{\text{mkt}}(K_j, T_j) \right]^2, \quad (13)$$

where

$$w_j = (0.05 + |K_j/S_0 - 1|)^{-1}. \quad (14)$$

This gives more importance to near-the-money options, which are central for hedging and pricing, while still penalising errors in the wings.

For each candidate parameter vector $(v_0, \kappa, \theta, \sigma, \rho)$, the calibration loop proceeds as follows:

1. Compute Heston model prices using the semi-analytical characteristic-function formula in equation (11).
2. Convert each Heston price into a Black-Scholes implied volatility by numerical inversion.
3. Compare model implied volatilities with market implied volatilities using the loss function in equation (13).

This approach is deterministic: the same parameter vector always produces the same objective value. This is preferable to Monte Carlo pricing inside the calibration loop, which would introduce simulation noise and slow convergence.

5.3 Optimisation Algorithm

The objective function (13) is non-convex and may have multiple local minima. Gradient-based methods such as L-BFGS-B or SLSQP can be sensitive to the starting point. For this reason, the implementation uses `differential_evolution` from `scipy.optimize` [10], a population-based global optimisation algorithm based on mutation and crossover.

The parameter bounds are:

Parameter	Lower bound	Upper bound
v_0	10^{-4}	0.50
κ	0.05	10.0
θ	10^{-4}	0.50
σ	10^{-3}	3.00
ρ	-0.95	-0.20

The restriction $\rho < 0$ reflects the standard equity-index leverage effect. The upper bound $\rho = -0.20$ avoids values close to zero, which would effectively remove the main mechanism that generates skew. The final calibration uses 50 generations, a population size of 6 per parameter dimension, and a polishing step.

5.4 Volatility Surface Reconstruction

After calibration, the Heston volatility surface is reconstructed by pricing puts on a regular grid of moneyness and maturity using the same characteristic-function pricer. These model prices are then converted into Black-Scholes implied volatilities. The market surface is obtained by interpolating the observed put implied volatilities on the same grid. This allows a direct comparison between the market surface, the Heston surface, and the absolute fitting error.

6 Results

6.1 Calibrated Parameters

Table 2 reports the calibrated Heston parameters.

Table 2: Calibrated Heston parameters for SPX puts, excluding maturities below 0.02 years.

Parameter	Value	Interpretation	Note
v_0	0.0310	Initial vol = 17.6%	Moderate initial variance
κ	0.1737	Slow mean reversion	Half-life ≈ 4 years
θ	0.0145	Long-run vol = 12.1%	Low long-run variance
σ	0.8363	High vol-of-vol	Flexible variance dynamics
ρ	-0.200	Negative correlation	Upper edge of imposed range
RMSE (IV)		4.78%	
Feller condition		Violated (-0.694)	Common in practice

The estimated initial volatility is $\sqrt{v_0} = 17.6\%$, which is a plausible short-horizon volatility level for the SPX on the data date. The low value of κ implies slow mean reversion: the

half-life of a variance shock is approximately $\ln(2)/\kappa \approx 4$ years. Because the calibration uses only short maturities, κ and θ are less strongly identified than v_0 , σ , and ρ .

The calibrated correlation is negative, as expected for equity-index options. However, it lies at the upper edge of the imposed interval, $\rho = -0.20$. This suggests that negative spot-volatility correlation is still required to generate skew, but the exact value of ρ is weakly identified in this short-maturity dataset. The Feller condition is violated, which is common in empirical Heston calibrations and is therefore reported rather than imposed.

6.2 Implied Volatility Smiles

Figure 1 compares the market and Heston implied volatility curves for each retained maturity.

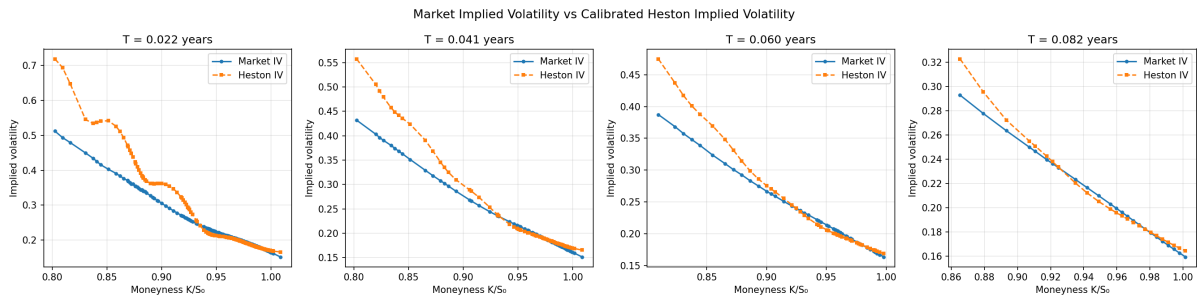


Figure 1: Market implied volatility (blue) versus calibrated Heston implied volatility (orange) for four maturities. Moneyness is defined as K/S_0 .

The model captures the broad downward-sloping skew, especially for the longer maturities and the near-the-money region. For the shortest retained maturity, $T = 0.022$ years, the Heston curve overestimates implied volatility for deep out-of-the-money puts. This is consistent with the fact that very short-dated equity-index smiles are often driven by jump-like tail risk and demand for crash protection, while the standard Heston model is a continuous diffusion.

6.3 Volatility Surface Comparison

Figure 2 shows the three-dimensional comparison between the market and Heston implied volatility surfaces, together with the absolute fitting error.

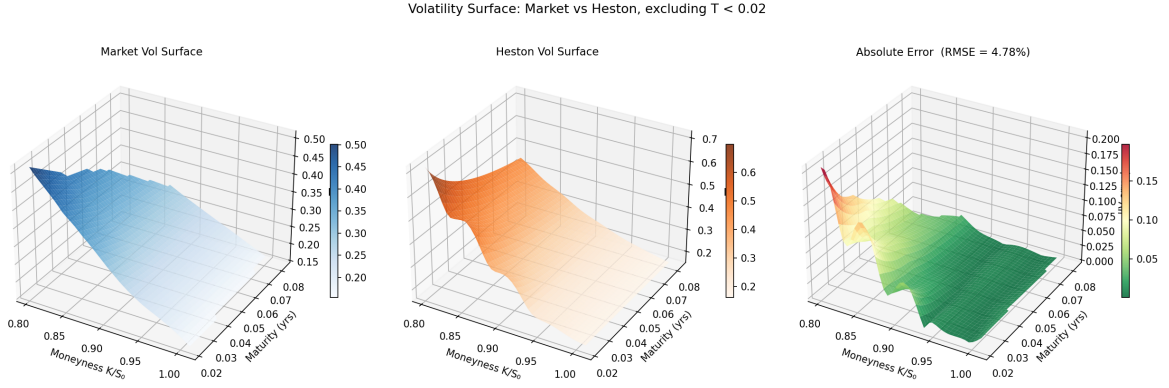


Figure 2: Left: market implied volatility surface interpolated from put quotes. Centre: Heston model surface computed with the characteristic function pricer. Right: absolute fitting error. $RMSE = 4.78\%$.

Both surfaces display the same qualitative structure: implied volatility increases as moneyness decreases, producing a left skew, and tends to decline with maturity. The fitting error is concentrated in the short-maturity, low-moneyness region, which is also where the market surface is steepest. Across much of the surface, the absolute error remains moderate, suggesting that the calibrated Heston model provides a reasonable practitioner-level approximation despite its structural limitations.

7 Conclusion

This project calibrated the Heston stochastic volatility model to S&P 500 put options using real market data and evaluated the quality of the fit through both smile-level and surface-level comparisons.

Main findings. The calibrated model achieves an implied-volatility RMSE of 4.78%. The fit is best in the near-the-money region and for the longer retained maturities. The parameters are economically interpretable: the model produces a moderate initial volatility, high vol-of-vol, and negative spot-volatility correlation, which together generate a downward-sloping equity-index skew. The model struggles most in the short-maturity, deep-out-of-the-money region, where the market prices crash protection more aggressively than a pure diffusion model can reproduce.

Limitations. Three main limitations should be noted. First, the standard Heston model is a continuous diffusion and therefore cannot capture sudden jumps in the asset price or variance, which are important at very short maturities. Second, the risk-free rate is proxied by a single 3-month Treasury bill yield and no dividend or forward-price adjustment is made, which can introduce a small pricing mismatch for index options. Third, the

calibration relies only on short-dated options, so long-run parameters such as κ and θ are not strongly identified. In addition, although the characteristic-function pricer is deterministic, the results still depend on the chosen integration range and discretisation accuracy.

Further work. Natural extensions include: (i) using a maturity-dependent interest-rate and dividend or forward curve; (ii) adding jumps to either the price process (Bates, 1996) or the variance process (Duffie et al., 2000) to better fit short-maturity smiles; and (iii) exploring the rough Heston model [5], in which the variance process is driven by a fractional Brownian motion with Hurst exponent $H < 1/2$, which has been shown to provide superior fits to the full volatility surface.

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